

Carrier marginal-sensitivity

Marginal-robust

Marginal-sensitive

Shift symmetry

Class-
symmetric

**DA NEUTRAL
/ HURTS**

LR, MLP

S0 empirical

DA HELPS

*LightGBM,
RF, ExtraTrees*

S0 empirical

Class-
asymmetric

**DA HELPS
(predicted)**

S2/S3 pending

**DA HELPS
STRONGLY
(predicted)**

S2/S3 pending